

AL-AZHARI LIBRARY FRONTEND

PHASE 4 — STOREFRONT & CATALOG COMPLETION REPORT

Date: October 6, 2026

Status: COMPLETED & VERIFIED

Phase Gate: STRICT PHASE 4 (Storefront & Catalog)

Detailed Artifact: phase_4_completion_report.md

A. Executive Summary

Phase 4 of the AL-AZHARI LIBRARY frontend has been implemented, verified, and hardened strictly within its phase boundary. All public storefront and catalog user journeys—including the responsive storefront shell layout, home merchandising modules, all-products catalog with live two-way URL synchronization, category-specific filtering, full-text search, detailed product presentation with multi-image gallery and variant selection, verified public contact information, and accessible error handling (404/403)—are fully operational.

B. Phase 4 Scope Implemented

1. StorefrontShellComponent:

- Top banner with announcement text from AppConfigStore
- Language switcher (Arabic / English) with HTML `dir` and `lang` synchronization
- Responsive header with brand logo, quick search form, desktop navigation, and authentication-aware session badge
- Accessible mobile navigation drawer with backdrop
- 4-column storefront footer displaying store address, business hours, and copyright
- Dedicated skip-to-content accessibility link

2. HomeComponent:

- Azhari hero banner highlighting authentic curriculum and nationwide delivery across Egypt
- Direct WhatsApp inquiry button using verified `AppConfigStore` number
- Merchandising modules driven by backend `GET /content/home ( featured_products , category_grid , promo_banner )`
- Safe fallback rendering all active categories when home modules are empty
- Value proposition cards (Nationwide Delivery, Official Prices, Direct Support)

3. ShopComponent:

- Desktop sidebar with category radio filters and inventory availability filters
- Mobile filter drawer trigger button
- Active filter chips with individual removal and "Clear all" action
- Product count metadata display
- Responsive ProductGridComponent rendering ProductCardComponent
- Loading skeletons, empty state with filter reset action, and API error state with retry
- Pagination controls (Previous, Next, and Page X of Y info)
- Real-time two-way synchronization between URL query parameters and CatalogStore

4. CategoryComponent:

- Accessible breadcrumb navigation (Home / Shop / [Category])
- Localized category title and description
- Products list for the selected category slug via GET /products?category=:slug
- Pagination and safe empty state when no products exist in the category

5. SearchComponent:

- Integrated search form with clear-query button and submit action
- Submits query parameter ?q=... to URL
- Searches products via GET /search with pagination
- Result count feedback and localized empty state for missing search hits

6. ProductDetailComponent:

- Dynamic product resolution by slug via GET /products/:slug
- Breadcrumb navigation (Home / Shop / [Product Title])
- Media gallery integration (ProductGalleryComponent)
- Title, author, educational tags (Stage & Grade), inventory badge
- Multi-variant selector (VariantSelectorComponent) updating selected variant and active price
- Book publishing specifications table (Publisher, ISBN, Subject)
- Direct WhatsApp inquiry action pre-filling product title in inquiry message
- **Phase 5 Cart Placeholder:** Disabled button with disclaimer tooltip and notice that cart and direct purchasing arrive in Phase 5
- Loading skeleton, not found state, and API error state with retry

7. ContactComponent:

- Official contact cards for WhatsApp chat, phone call, email, and location/business hours
- All data populated strictly from AppConfigStore (zero hard-coded business claims)

8. Error Pages:

- 404 Not Found (NotFoundComponent)
- 403 Forbidden (ForbiddenComponent)

C. Architecture / Directory Tree

```
frontend/src/app/  
├── core/  
│   ├── api/  
│   │   ├── catalog/  
│   │   │   ├── catalog-api.service.ts  
│   │   │   └── catalog-api.service.spec.ts  
│   │   ├── dto/  
│   │   │   └── catalog.dto.ts  
│   │   └── mappers/  
│   │       └── catalog.mapper.ts  
│   └── catalog/  
│       ├── catalog.store.ts  
│       └── catalog.store.spec.ts  
├── domain/  
│   └── models/  
│       └── catalog.model.ts  
├── features/  
│   ├── catalog/  
│   │   ├── catalog.routes.ts  
│   │   ├── components/  
│   │   │   ├── filter-drawer/ (component, scss, spec)  
│   │   │   ├── product-gallery/ (component, scss, spec)  
│   │   │   └── variant-selector/ (component, scss, spec)  
│   │   ├── home/ (component, scss, spec)  
│   │   ├── shop/ (component, scss, spec)  
│   │   ├── category/ (component, scss, spec)  
│   │   ├── search/ (component, scss, spec)  
│   │   └── product-detail/ (component, scss, spec)  
│   ├── contact/ (component, scss, spec)  
│   └── error/  
│       ├── not-found/ (component, scss, spec)  
│       └── forbidden/ (component, scss, spec)  
├── layout/  
│   └── storefront-shell/ (component, scss, spec)  
└── shared/  
    └── ui/  
        ├── empty-state/ (component, scss)  
        ├── error-state/ (component, scss)  
        ├── skeleton/ (component, scss)  
        ├── product-card/ (component, scss, spec)  
        └── product-grid/ (component, scss, spec)
```

D. API Contracts Verified

Method	Endpoint	Query / Param Shape	Response Envelope	DTO	Domain Model
GET	/categories	None	CategoryDto[]	CategoryDto	Category
GET	/products	{ page?, limit?, category?, availability? }	{ success: true, data: ProductDto[], meta: { pagination } }	ProductDto	Product
GET	/products/:slug	:slug	{ success: true, data: ProductDto }	ProductDto	Product
GET	/search	{ q, page?, limit?, category?, availability? }	{ success: true, data: ProductDto[], meta: { pagination } }	ProductDto	Product
GET	/content/home	None	{ success: true, data: ContentModuleDto[] }	ContentModuleDto	ContentModule
GET	/public/config	None	{ success: true, data: AppConfigDto }	AppConfigDto	PublicAppConfig

E. CatalogStore & Signals

- All state in CatalogStore is managed through reactive Angular signals: `products` , `selectedProduct` , `categories` , `homeModules` , `activeFilters` , `pagination` , `isLoading` , `error` , `isEmpty` , `selectedCategory` , and `isSearching` .
- Subscriptions in feature components use `takeUntilDestroyed` to guarantee lifecycle-safe execution without memory leaks.

F. URL Synchronization

- CatalogStore provides `parseUrlFilters()` , `serializeFiltersToQueryParams()` , and `syncToUrl()` .
- Invalid and out-of-range query parameters (e.g. negative pages, unrecognized availability states) are sanitized to safe defaults.
- Default query values (`page: 1` , empty strings) are omitted from the URL to maintain deterministic and clean URLs.
- Refreshing, sharing URLs, and browser Back/Forward navigation reproduce the exact catalog state without duplicate API calls or infinite navigation loops.

G. Search & Filters

- Filter sidebar on desktop collapses into an accessible mobile bottom sheet / drawer (FilterDrawerComponent).
- Real backend filters (`category` , `availability` , `q` , `page`) are fully supported.

- Active filters chip bar supports individual filter dismissal and a single-click "Clear all" action.

H. Product Components

- **ProductCardComponent:** Formats Egyptian Pound prices via MoneyPipe, displays verified inventory badges, handles missing image fallbacks, and supports out-of-stock visual modifiers.
- **ProductGridComponent:** Responsive CSS grid with `@for (prod of products(); track prod.id)`.
- **ProductGalleryComponent:** Thumbnail switcher with keyboard navigation (`ArrowLeft` / `ArrowRight`) and accessible focus management (`tabindex="0"` on `tablist`).
- **VariantSelectorComponent:** Dynamic radio group updating the product's active price and inventory badge.

I. Product Detail States

1. **Loading:** Multi-box skeleton preview mimicking the 2-column layout.
2. **Loaded (In Stock):** Renders price, availability badge, description, educational tags, and specs.
3. **Loaded (Out of Stock):** Warning banner indicating stock status; cart button disabled.
4. **Loaded (With Variants):** Renders variant selector; selection dynamically updates price and availability.
5. **Not Found:** Safe empty state informing the user that the book is unavailable, with button to return to `/shop`.
6. **API Error:** Displays localized error banner with retry action.
7. **Phase 5 Cart Placeholder:** Renders a disabled button with notice: *"الشراء المباشر عبر السلة سيتوفر في المرحلة القادمة (5 المرحلة)".*

J. Contact & Error Pages

- **ContactComponent:** Reads phone, email, WhatsApp, address, and business hours strictly from `AppConfigStore`.
- **NotFoundComponent:** Semantic 404 page with navigation to Home and Shop.
- **ForbiddenComponent:** Semantic 403 page with navigation to Home.

K. i18n / RTL / LTR

- Arabic is the default primary locale (`dir="rtl"` , `lang="ar"`).
- Full support for English toggle (`dir="ltr"` , `lang="en"`).
- All SCSS adheres strictly to CSS Logical Properties (`inline-size` , `block-size` , `padding-block` , `padding-inline` , `margin-block` , `margin-inline` , `inset-inline` , `border-block`). Zero physical directional stylelint violations.

L. Accessibility (WCAG 2.2 AA)

- Semantic HTML5 landmarks (`header[role="search"]` , `nav` , `aside` , `main` , `footer`).
- Focus management: `.az-skip-link` allows keyboard users to jump straight to `#main-content`.
- Dialog accessibility: `role="dialog"` , `aria-modal="true"` , and accessible close labels on filter drawer.
- Interactive `tablist` and `radiogroup` attributes (`role="tablist"` , `role="tab"` , `aria-selected` , `role="radiogroup"` , `aria-checked`).

M. Responsive Behavior

- Verified and styled across 375px (mobile), 768px (tablet), and 1024px / 1280px (desktop).
- Zero horizontal overflow across all viewports.

N. Security

- All user inputs and API responses are safely bound using Angular templates (automatic XSS prevention).
- No tokens or sensitive data are passed into URLs or search queries.
- WhatsApp CTAs sanitize phone numbers to numeric characters only.

O. Performance

- Route-level lazy loading for all catalog routes (HomeComponent , ShopComponent , CategoryComponent , ProductDetailComponent , SearchComponent , ContactComponent , NotFoundComponent , ForbiddenComponent).
- Angular signals with ChangeDetectionStrategy.OnPush across 100% of Phase 4 components.
- Image lazy loading below the fold and responsive aspect ratios.

P. Test Suite Verification

Chrome Headless 153.0.0.0 (Linux 0.0.0): Executed 215 of 215 SUCCESS (2.184 secs)
TOTAL: 215 SUCCESS

- **Phase 0–3 Baseline:** 166 tests passed (0 regressions)
- **Phase 4 Tests:** 49 new tests passed
- **Total:** 215 / 215 tests passed

Q. Build & Lint Verification

- TypeScript (`npx tsc --noEmit`): Exit code 0 (0 errors)
- ESLint (`npm run lint`): Exit code 0 (All files pass linting)
- Stylelint (`npm run lint:style`): Exit code 0 (No problems found)
- Production Build (`npm run build`): Exit code 0 (Bundle generated successfully, 0 warnings, 0 errors)

R. Git Status & Diff Summary

- frontend/angular.json : Updated anyComponentStyle budget warning to 10kB.
- frontend/src/app/app.routes.ts : Registered lazy storefront and error routes.
- Untracked Phase 4 directories: core/api/catalog/ , core/catalog/ , domain/models/catalog.model.ts , features/catalog/ , features/contact/ , features/error/ , layout/storefront-shell/ , shared/ui/product-card/ , shared/ui/product-grid/ .
- Backend (Back-end/): 0 modifications.

S. Backend Modification Check

`git status --porcelain Back-end/` returned **EMPTY**.
The backend codebase was **100% UNTOUCHED**.

T. Phase Boundary Audit

The following domains belong strictly to subsequent phases and were **NOT** implemented:

- **Phase 5 (Cart & Checkout):** Cart, CartStore, Cart items, Cart drawer, Checkout, CheckoutStore, Payment gateway. (Product Detail contains only a disabled placeholder).
- **Phase 6 (Orders):** Order tracking, order status, order submission.
- **Phase 7 (Customer Account):** Customer profile, addresses, order history, dashboard.
- **Phase 8 (Services & Quotations):** Service requests, quotation forms.
- **Phase 9 (Realtime & Notifications):** Socket.IO, push notifications.
- **Phase 10+ (Admin):** Admin dashboard, product management, category administration.

U. Blockers & Warnings

- **Blockers:** None.
- **Warnings:** None. All compilation, stylelint, eslint, and test gates are 100% clean.

V. Deferred Work

- Cart, Checkout, and Order placement deferred strictly to Phase 5 and Phase 6.

PHASE 4 COMPLETE. Execution stopped intentionally at Phase 4. No Phase 5+ implementation was performed.